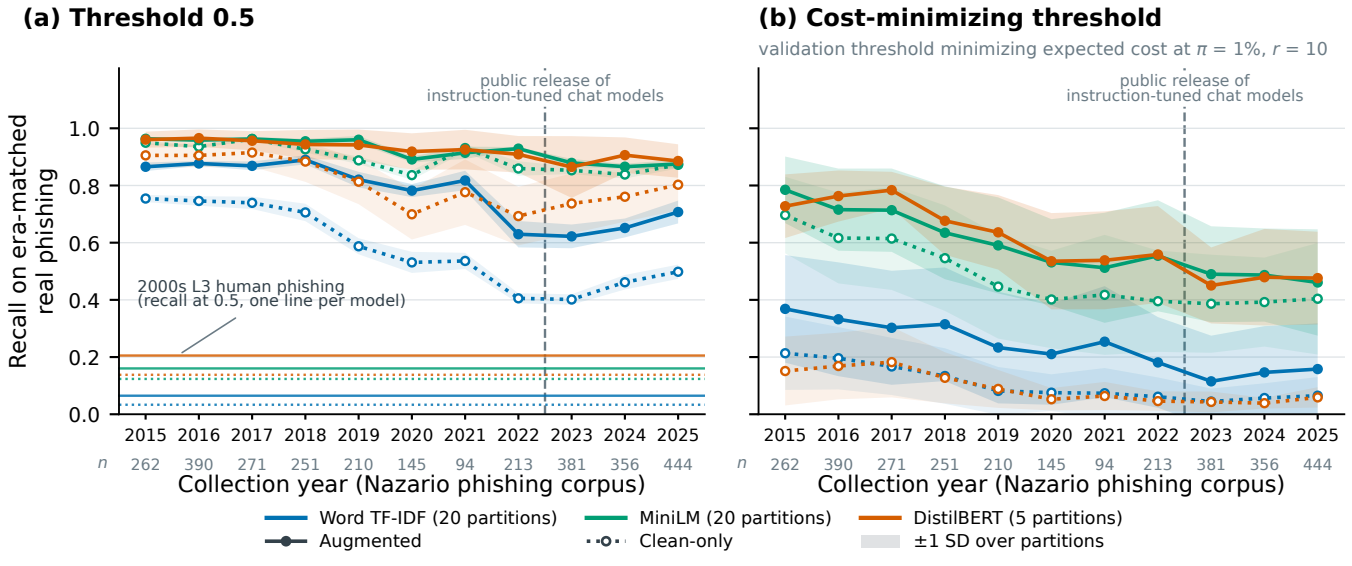
(a) Threshold 0.5
(b) Cost-minimizing threshold
validation threshold minimizing expected cost at π = 1%, r = 10
public release of instruction-tuned chat models
Recall on era-matched real phishing
2000s L3 human phishing (recall at 0.5, one line per model)
1.0
0.8
0.6
0.4
0.2
0.0
2015 2016 2017 2018 2019 2020 2021 2022 2023 2024 2025
n 262 390 271 251 210 145 94 213 381 356 444
Collection year (Nazario phishing corpus)
Word TF-IDF (20 partitions)
MiniLM (20 partitions)
DistilBERT (5 partitions)
Augmented
Clean-only
±1 SD over partitions